

FinSpark Hackathon - Problem Statement 2

Strategy Document

Problem Statement 2: AI-Driven Correlation of Cybersecurity Telemetry & Transactional Behaviour

Objective

Develop an AI-powered platform that correlates cybersecurity telemetry (VPN logs, firewall events, device information, authentication logs, email alerts, SIEM alerts) with banking transactional behaviour. Instead of detecting isolated frauds, the platform builds contextual intelligence across multiple signals, allowing the bank to identify account takeover, insider threats, mule accounts, coordinated fraud, and emerging cyber attacks with explainable AI.

Proposed Solution (Deep Overview)

1. Data Collection

Collect transaction records, login events, device fingerprints, IP reputation, employee activities, ATM usage, UPI logs and threat intelligence feeds.

2. Data Correlation Engine

Convert all events into a unified timeline and correlate users, devices, IPs, merchants and transactions.

3. AI Risk Engine

Use anomaly detection, supervised fraud models and behavioural profiling to calculate a dynamic risk score.

4. Explainable AI

Generate human-readable explanations for every alert so analysts understand why the model flagged it.

5. Knowledge Graph

Visualize relationships between customers, employees, devices, merchants and suspicious entities.

6. Executive Dashboard

Provide SOC analysts and bank executives with real-time alerts, trends, incident timelines and recommendations.

Top 10 Innovations

1. Digital DNA

Create a behavioural fingerprint for every customer and employee based on transaction timing, location, device usage and spending habits. Alerts are triggered when behaviour deviates significantly.

2. Attack Story Generator

Automatically reconstruct the complete attack timeline instead of showing isolated alerts.

3. AI Investigator

LLM-generated investigation summary explaining why an event is suspicious and recommended actions.

4. Predictive Threat Engine

Predict possible account takeover or fraud before money is transferred.

5. Behaviour Twin

Maintain a virtual behavioural model of every user and compare live actions against it.

6. Knowledge Graph

Graph-based visualization connecting users, devices, IPs, merchants and transactions.

7. Dynamic Risk Timeline

Risk score evolves in real time as new events occur.

8. Trust Score Engine

Assign trust scores to customer, device, merchant and IP before approving transactions.

9. Quantum Readiness Monitor

Highlight systems using legacy cryptography and estimate future quantum exposure.

10. Multi-Agent AI

Separate AI agents for cyber analysis, fraud detection, behavioural analytics, risk prediction and executive reporting.

Comparison: Problem Statement 1 vs Problem Statement 2

Evaluation Criteria	PS1	PS2	Winner
Business Relevance (40%)	High	Very High	PS2
Security (30%)	Excellent	Excellent	Tie
Innovation (15%)	Moderate	Very High	PS2
User Experience (5%)	Good	Very Good	PS2
Scalability (5%)	High	Very High	PS2
Ease of Development (5%)	Moderate	High	PS2
Overall Recommendation	Strong	Outstanding	PS2

Conclusion: Based on the evaluation criteria, Problem Statement 2 offers stronger business impact, greater innovation potential, and a more impressive prototype opportunity for the FinSpark Hackathon.